EXPNO:2 DATE:	C Compiler in Virtual Machine

Aim:

Install a C compiler (GCC) in the Linux VM and execute simple programs.

Pre-Requisites:

- Windows 10 Host Machine
- Oracle VM VirtualBox (Open Source)
- Ubuntu Desktop ISO image (Open Source)

Procedure:

Step 1: Open Terminal

In the Ubuntu VM, press Ctrl+Alt+T to open the terminal.

Step 2: Update PackageRepository

```
sudo apt update
```

Step 3: Install Build Essentials

The build-essential package includes the GCC compiler and other necessary tools
.

```
sudo aptinstallbuild-essential-y
```

Step 4: Verify Installation

```
gcc --version
```

Step 5: Write a Simple C Program

Create a file named hello.c using a text editor like nano:

```
nano hello.c
```

Enter the following code:

```
#include  <stdio.h>
int main() {
    printf("Hello, Virtual Cloud World!\n");
    return 0;
}
```

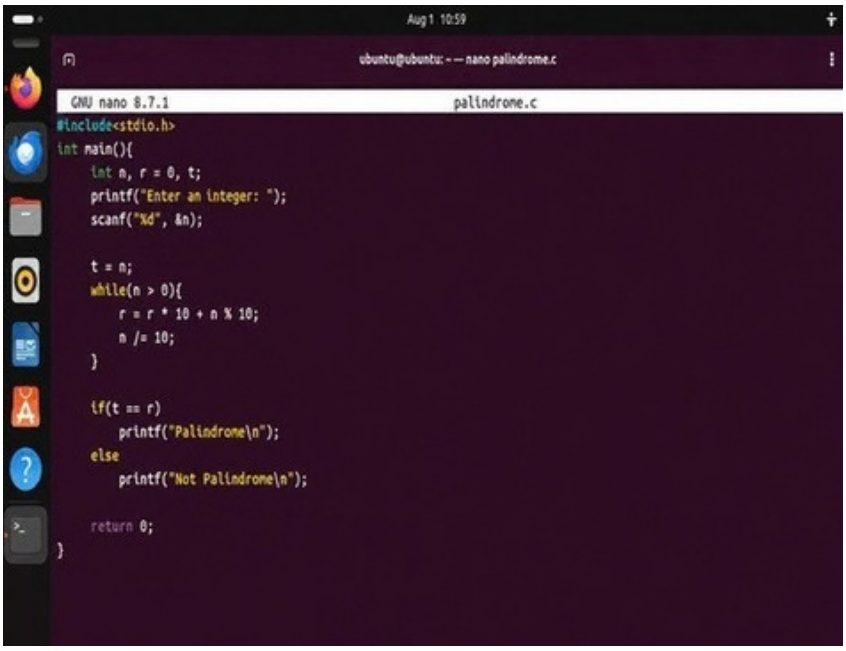
Step6:CompileandExecute

```
gcc hello.c -o hello
./hello
```

Output should display: Hello, Virtual Cloud World!

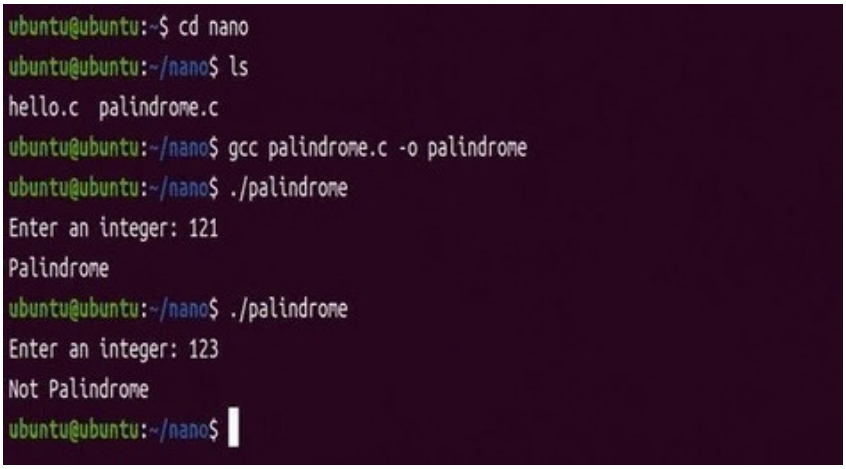
Sample Program: Palindrome Check:

A second program, palindrome.c, was written to check whether a given integer is a palindrome:



Output:

The palindrome.c program was compiled and executed, and tested with two inputs (121 and 123):



Result:

Thus,theGCC compiler was successfully installed in the Linux VM and used to compile and run C programs.